# STT 843: Multivariate Analysis

## 9. Canonical Variates and Canonical Correlation (Chapter 10.5)

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# Outline

1. Additional Sample Descriptive Measures
   - Large Sample Inferences

# Additional Sample Descriptive Measures

- Matrices of Errors of Approximations: Given the matrices $\hat{\mathbf{A}}$ and $\hat{\mathbf{B}}$ defined in (10-27), let $\hat{\mathbf{a}}^{(i)}$ and $\hat{\mathbf{b}}^{(i)}$ denote the ith column of $\hat{\mathbf{A}}^{-1}$ and $\hat{\mathbf{B}}^{-1}$, respectively.
- Since $\hat{\mathbf{U}} = \hat{\mathbf{A}}\mathbf{x}^{(1)}$ and $\hat{\mathbf{V}} = \hat{\mathbf{B}}\mathbf{x}^{(2)}$ we can write

$$\underset{(p\times 1)}{\mathbf{x}^{(1)}} = \underset{(p\times p)}{\hat{\mathbf{A}}^{-1}}\underset{(p\times 1)}{\hat{\mathbf{U}}} \qquad \underset{(q\times 1)}{\mathbf{x}^{(2)}} = \underset{(q\times q)}{\hat{\mathbf{B}}^{-1}}\underset{(q\times 1)}{\hat{\mathbf{V}}}$$

Because sample $\mathrm{Cov}(\hat{\mathbf{U}}, \hat{\mathbf{V}}) = \hat{\mathbf{A}}\mathbf{S}_{12}\hat{\mathbf{B}}^{\mathsf{T}}$, sample $\mathrm{Cov}(\hat{\mathbf{U}}) = \hat{\mathbf{A}}\mathbf{S}_{11}\hat{\mathbf{A}}^{\mathsf{T}} = \underset{(p\times p)}{\mathbf{I}}$, and $q_i$ sample $\mathrm{Cov}(\hat{\mathbf{V}}) = \hat{\mathbf{B}}\mathbf{S}_{22}\hat{\mathbf{B}}^{\mathsf{T}} = \underset{(q\times q)^{\mathsf{T}}}{\mathbf{I}}$

$$\mathbf{S}_{11} = \left(\hat{\mathbf{A}}^{-1}\right)\left(\hat{\mathbf{A}}^{-1}\right)^{\mathsf{T}} = \hat{\mathbf{a}}^{(1)}\hat{\mathbf{a}}^{(1)\,\mathsf{T}} + \hat{\mathbf{a}}^{(2)}\hat{\mathbf{a}}^{(2)\,\mathsf{T}} + \cdots + \hat{\mathbf{a}}^{(p)}\hat{\mathbf{a}}^{(p)\,\mathsf{T}}$$

$$\mathbf{S}_{22} = \left(\hat{\mathbf{B}}^{-1}\right)\left(\hat{\mathbf{B}}^{-1}\right)^{\mathsf{T}} = \hat{\mathbf{b}}^{(1)}\hat{\mathbf{b}}^{(1)\,\mathsf{T}} + \hat{\mathbf{b}}^{(2)}\hat{\mathbf{b}}^{(2)\,\mathsf{T}} + \cdots + \hat{\mathbf{b}}^{(q)}\hat{\mathbf{b}}^{(q)\,\mathsf{T}}$$

- Since $\mathbf{x}^{(1)} = \hat{A}^{-1}\hat{\mathbf{U}}$ and $\hat{\mathbf{U}}$ has sample covariance $\mathbf{I}$, the first $r$ columns contain the sample covariances of the first $r$ canonical variates $\hat{U}_1, \hat{U}_2, \ldots,$ their component variables $X_1^{(1)}, X_2^{(1)}, \ldots, X_p^{(1)}$.
- the first $r$ columns contain the sample covariance of $\hat{V}_1, \hat{V}_2, \ldots, \hat{V}_r$ with their component variables

If only the first $r$ canonical pairs are used, so that for instance,

$$\widetilde{\mathbf{x}}^{(1)} = \begin{bmatrix} \hat{\mathbf{a}}^{(1)} & \hat{\mathbf{a}}^{(2)} & \cdots & \hat{\mathbf{a}}^{(r)} \end{bmatrix} \begin{bmatrix} \hat{U}_1 \\ \hat{U}_2 \\ \vdots \\ \hat{U}_r \end{bmatrix}$$

and

$$\widetilde{\mathbf{x}}^{(2)} = \begin{bmatrix} \hat{\mathbf{b}}^{(1)} & \hat{\mathbf{b}}^{(2)} & \cdots & \hat{\mathbf{b}}^{(r)} \end{bmatrix} \begin{bmatrix} \hat{V}_1 \\ \hat{V}_2 \\ \vdots \\ \hat{V}_r \end{bmatrix}$$

then $\mathbf{S}_{12}$ is approximated by sample Cov $\left(\widetilde{\mathbf{x}}^{(1)}, \widetilde{\mathbf{x}}^{(2)}\right)$.

Continuing, we see that the matrices of errors of approximation are

$$\mathbf{S}_{11} - \left(\hat{\mathbf{a}}^{(1)}\hat{\mathbf{a}}^{(1)\,\mathsf{T}} + \hat{\mathbf{a}}^{(2)}\hat{\mathbf{a}}^{(2)\,\mathsf{T}} + \cdots + \hat{\mathbf{a}}^{(r)}\hat{\mathbf{a}}^{(r)\,\mathsf{T}}\right)$$
$$= \hat{\mathbf{a}}^{(r+1)}\hat{\mathbf{a}}^{(r+1)\,\mathsf{T}} + \cdots + \hat{\mathbf{a}}^{(p)}\hat{\mathbf{a}}^{(p)\,\mathsf{T}}$$
$$\mathbf{S}_{22} - \left(\hat{\mathbf{b}}^{(1)}\hat{\mathbf{b}}^{(1)\,\mathsf{T}} + \hat{\mathbf{b}}^{(2)}\hat{\mathbf{b}}^{(2)\,\mathsf{T}} + \cdots + \hat{\mathbf{b}}^{(r)}\hat{\mathbf{b}}^{(r)\,\mathsf{T}}\right)$$
$$= \hat{\mathbf{b}}^{(r+1)}\hat{\mathbf{b}}^{(r+1)\,\mathsf{T}} + \cdots + \hat{\mathbf{b}}^{(q)}\hat{\mathbf{b}}^{(q)\,\mathsf{T}}$$
$$\mathbf{S}_{12} - \left(\widehat{\rho_1^*}\hat{\mathbf{a}}^{(1)}\hat{\mathbf{b}}^{(1)\,\mathsf{T}} + \widehat{\rho_2^*}\hat{\mathbf{a}}^{(2)}\hat{\mathbf{b}}^{(2)\,\mathsf{T}} + \cdots + \widehat{\rho_r^*}\hat{\mathbf{a}}^{(r)}\hat{\mathbf{b}}^{(r)\,\mathsf{T}}\right)$$
$$= \widehat{\rho_{r+1}^*}\hat{\mathbf{a}}^{(r+1)}\hat{\mathbf{b}}^{(r+1)\,\mathsf{T}} + \cdots + \widehat{\rho_p^*}\hat{\mathbf{a}}^{(p)}\hat{\mathbf{b}}^{(p)\,\mathsf{T}}$$

- The approximation error matrices may be interpreted as descriptive summaries of how well the first $r$ sample canonical variates reproduce the sample covariance matrices. Patterns of large entries in the rows and/or columns of the approximation error matrices indicate a poor "fit" to the corresponding variable(s).
- Ordinarily, the first $r$ variates do a better job of reproducing the elements of $\mathbf{S}_{12} = \mathbf{S}_{21}^{\mathsf{T}}$ than the elements of $\mathbf{S}_{11}$ or $\mathbf{S}_{22}$.

- Mathematically, this occurs because the residual matrix in the former case is directly related to the smallest $p - r$ sample canonical correlations. These correlations are usually all close to zero.
- On the other hand, the residual matrices associated with the approximations to the matrices $\mathbf{S}_{11}$ and $\mathbf{S}_{22}$ depend only on the last $p - r$ and $q - r$ coefficient vectors. The elements in these vectors may be relatively large, and hence, the residual matrices can have "large" entries.
- For standardized observations, $\mathbf{R}_{kl}$ replaces $\mathbf{S}_{kl}$ and $\hat{\mathbf{a}}_{\mathbf{z}}^{(k)}, \hat{\mathbf{b}}_{\mathbf{z}}^{(l)}$ replace $\hat{\mathbf{a}}^{(k)}, \hat{\mathbf{b}}^{(l)}$ in (10-34).

# Example 10.6 (Calculating matrices of errors of approximation)

In Example 10.4, we obtained the canonical correlations between the two head and the two leg variables for white leghorn fowl. Starting with the sample correlation matrix

$$\mathbf{R} = \begin{bmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{R}_{21} & \mathbf{R}_{22} \end{bmatrix} = \begin{bmatrix} 1.0 & .505 & .569 & .602 \\ .505 & 1.0 & .422 & .467 \\ .569 & .422 & 1.0 & .926 \\ .602 & .467 & .926 & 1.0 \end{bmatrix}$$

we obtained the two sets of canonical correlations and variables

$$\widehat{\rho_1^*} = .631 \quad \begin{aligned} \hat{U}_1 &= .781 z_1^{(1)} + .345 z_2^{(1)} \\ \hat{V}_1 &= .060 z_1^{(2)} + .944 z_2^{(2)} \end{aligned}$$

and

$$\widehat{\rho_2^*} = .057 \quad \begin{aligned} \hat{U}_2 &= -.856 z_1^{(1)} + 1.106 z_2^{(1)} \\ \hat{V}_2 &= -2.648 z_1^{(2)} + 2.475 z_2^{(2)} \end{aligned}$$

where $z_i^{(1)}, i = 1, 2$ and $z_i^{(2)}, i = 1, 2$ are the standardized data values for sets 1 and 2 , respectively.

We first calculate (see Panel 10.1)

$$\hat{\mathbf{A}}_{\mathbf{z}}^{-1} = \begin{bmatrix} .781 & .345 \\ -.856 & 1.106 \end{bmatrix}^{-1} = \begin{bmatrix} .9548 & -.2974 \\ .7388 & .6739 \end{bmatrix}$$

$$\hat{\mathbf{B}}_{\mathbf{z}}^{-1} = \begin{bmatrix} .9343 & -.3564 \\ .9997 & .0227 \end{bmatrix}$$

Consequently, the matrices of errors of approximation created by using only the second canonical pair are

$$\mathbf{R}_{12} - \text{sample Cov}\left(\widetilde{\mathbf{z}}^{(1)}, \widetilde{\mathbf{z}}^{(2)}\right) = (.057)\begin{bmatrix} -.2974 \\ .6739 \end{bmatrix}\begin{bmatrix} -.3564 & .0227 \end{bmatrix}$$

$$= \begin{bmatrix} .006 & -.000 \\ -.014 & .001 \end{bmatrix}$$

$$\mathbf{R}_{11} - \text{sample Cov}\left(\widetilde{\mathbf{z}}^{(1)}\right) = \begin{bmatrix} -.2974 \\ .6739 \end{bmatrix}\begin{bmatrix} -.2974 & .6739 \end{bmatrix}$$

$$= \begin{bmatrix} .088 & -.200 \\ -.200 & .454 \end{bmatrix}$$

$$\mathbf{R}_{22} - \text{ sample Cov}\left(\widetilde{\mathbf{z}}^{(2)}\right) = \begin{bmatrix} -.3564 \\ .0227 \end{bmatrix}\begin{bmatrix} -.3564 & .0227 \end{bmatrix}$$

$$= \begin{bmatrix} .127 & -.008 \\ -.008 & .001 \end{bmatrix}$$

We see that the first pair of canonical variables effectively summarizes (reproduces) the intraset correlations in $\mathbf{R}_{12}$. However, the individual variates are not particularly effective summaries of the sampling variability in the original $z^{(1)}$ and $z^{(2)}$ sets, respectively. This is especially true for $\hat{U}_1$.

# Proportions of Explained Sample Variance

When the observations are standardized, the sample covariance matrices $\mathbf{S}_{kl}$ are correlation matrices $\mathbf{R}_{kl}$. The canonical coefficient vectors are the rows of the matrices $\hat{\mathbf{A}}_{\mathbf{z}}$ and $\hat{\mathbf{B}}_{\mathbf{z}}$ and the columns of $\hat{\mathbf{A}}_{\mathbf{z}}^{-1}$ and $\hat{\mathbf{B}}_{\mathbf{z}}^{-1}$ are the sample correlations between the canonical variates and their component variables.

Specifically,

$$\text{sample Cov}\left(z^{(1)}, \hat{\mathbf{U}}\right) = \text{sample Cov}\left(\hat{\mathbf{A}}_{\mathbf{z}}^{-1}\hat{\mathbf{U}}, \hat{\mathbf{U}}\right) = \hat{\mathbf{A}}_{\mathbf{z}}^{-1}$$

and

$$\text{sample Cov}\left(z^{(2)}, \hat{\mathbf{V}}\right) = \text{sample Cov}\left(\hat{\mathbf{B}}_{\mathbf{z}}^{-1}\hat{\mathbf{V}}, \hat{\mathbf{v}}\right) = \hat{\mathbf{B}}_{\mathbf{z}}^{-1}$$

so

$$\hat{\mathbf{A}}_{\mathbf{z}}^{-1} = \left[\hat{\mathbf{a}}_{\mathbf{z}}^{(1)}, \hat{\mathbf{a}}_{\mathbf{z}}^{(2)}, \ldots, \hat{\mathbf{a}}_{z}^{(\rho)}\right] = \begin{bmatrix} r_{\hat{U}_1, z_1^{(1)}} & r_{\hat{U}_2, z_1^{(1)}} & \cdots & r_{\hat{U}_p, z_1^{(1)}} \\ r_{\hat{U}_1, z_2^{(1)}} & r_{\hat{U}_2, z_2^{(1)}} & \cdots & r_{\hat{U}_p, z_2^{(1)}} \\ \vdots & \vdots & \ddots & \vdots \\ r_{\hat{U}_1, z_p^{(1)}} & r_{\hat{U}_2, z_p^{(1)}} & \cdots & r_{\hat{U}_p, z_p^{(1)}} \end{bmatrix}$$

$$\hat{\mathbf{B}}_z^{-1} = \left[\hat{\mathbf{b}}_z^{(1)}, \hat{\mathbf{b}}_z^{(2)}, \ldots, \hat{\mathbf{b}}_z^{(q)}\right] = \begin{bmatrix} r_{\hat{V}_1, z_1^{(2)}} & r_{\hat{V}_2, z_1^{(2)}} & \cdots & r_{\hat{V}_q, z_1^{(2)}} \\ r_{\hat{V}_1, z_2^{(2)}} & r_{\hat{V}_2, z_2^{(2)}} & \cdots & r_{\hat{V}_q, z_z^{(2)}} \\ \vdots & \vdots & \ddots & \vdots \\ r_{\hat{V}_1, z_q^{(2)}} & r_{\hat{V}_2, z_q^{(2)}} & \cdots & r_{\hat{V}_q, z_q^{(2)}} \end{bmatrix}$$

where $r_{\hat{U}_i, z_k^{(1)}}$ and $r_{\hat{V}_i, z_i^{(i)}}$ are the sample correlation coefficients between the quantities with subscripts.

Total (standardized) sample variance in first set

$$= \operatorname{tr}\left(\mathbf{R}_{11}\right) = \operatorname{tr}\left(\hat{\mathbf{a}}_{\mathbf{z}}^{(1)}\hat{\mathbf{a}}_{\mathbf{z}}^{(1)\,\mathsf{T}} + \hat{\mathbf{a}}_{\mathbf{z}}^{(2)}\hat{\mathbf{a}}_{\mathbf{z}}^{(2)\,\mathsf{T}} + \cdots + \hat{\mathbf{a}}_{\mathbf{z}}^{(p)}\hat{\mathbf{a}}_{\mathbf{z}}^{(p)\,\mathsf{T}}\right) = p$$

Total (standardized) sample variance in second set

$$= \operatorname{tr}\left(\mathbf{R}_{22}\right) = \operatorname{tr}\left(\hat{\mathbf{b}}_{\mathbf{z}}^{(1)}\hat{\mathbf{b}}_{\mathbf{z}}^{(1)\,\mathsf{T}} + \hat{\mathbf{b}}_{\mathbf{z}}^{(2)}\hat{\mathbf{b}}_{\mathbf{z}}^{(2)\,\mathsf{T}} + \cdots + \hat{\mathbf{b}}_{\mathbf{z}}^{(q)}\hat{\mathbf{b}}_{\mathbf{z}}^{(q)\,\mathsf{T}\,\mathsf{T}}\right) = q$$

Since the correlations in the first $r < p$ columns of $\hat{\mathbf{A}}_{\mathbf{z}}^{-1}$ and $\hat{\mathbf{B}}_{\mathbf{z}}^{-1}$ involve only the sample canonical variates $\hat{U}_1, \hat{U}_2, \ldots, \hat{U}_r$ and $\hat{V}_1, \hat{V}_2, \ldots, \hat{V}_r$, respectively, we define the contributions of the first $r$ canonical variates to the total (standardized) sample variances as

$$\operatorname{tr}\left(\hat{\mathbf{a}}_{\mathbf{z}}^{(1)}\hat{\mathbf{a}}_{\mathbf{z}}^{(1)\,\mathsf{T}} + \hat{\mathbf{a}}_{\mathbf{z}}^{(2)}\hat{\mathbf{a}}_{\mathbf{z}}^{(2)\,\mathsf{T}} + \cdots + \hat{\mathbf{a}}_{\mathbf{z}}^{(r)}\hat{\mathbf{a}}_{\mathbf{z}}^{(r)\,\mathsf{T}}\right) = \sum_{i=1}^{r}\sum_{k=1}^{p} r^2_{\hat{U}_{i,z_k^{(1)}}}$$

and

$$\operatorname{tr}\left(\hat{\mathbf{b}}_{\mathbf{z}}^{(1)}\hat{\mathbf{b}}_{\mathbf{z}}^{(1)\,\mathsf{T}} + \hat{\mathbf{b}}_{\mathbf{z}}^{(2)}\hat{\mathbf{b}}_{\mathbf{z}}^{(2)\,\mathsf{T}} + \cdots + \hat{\mathbf{b}}_{\mathbf{z}}^{(r)}\hat{\mathbf{b}}_{\mathbf{z}}^{(r)\,\mathsf{T}}\right) = \sum_{i=1}^{r}\sum_{k=1}^{q} r^2_{\hat{V}_{i,z_k^{(2)}}}$$

The proportions of total (standardized) sample variances "explained by" the first canonical variates then become

$$R_{\mathbf{z}}^{2(1)} \mid \hat{U}_1, \hat{U}_2 \ldots, \hat{U}_r = \begin{pmatrix} \text{proportion of total standardized} \\ \text{sample variance in first set} \\ \text{explained by } \hat{U}_1, \hat{U}_2, \ldots, \hat{U}_r \end{pmatrix}$$

$$= \frac{\operatorname{tr}\left(\hat{\mathbf{a}}_{\mathbf{z}}^{(1)} \hat{\mathbf{a}}_{\mathbf{z}}^{(1),} + \cdots + \hat{\mathbf{a}}_{\mathbf{z}}^{(r)} \hat{\mathbf{a}}_{\mathbf{z}}^{(r)\,\mathsf{T}}\right)}{\operatorname{tr}\left(\mathbf{R}_{11}\right)}$$

$$= \frac{\sum_{i=1}^{r} \sum_{k=1}^{p} r^2_{\hat{U}_i, z_k^{(1)}}}{p}$$

and

$$R_{\mathbf{z}}^{2(2)} \mid \hat{V}_1, \hat{V}_2, \ldots, \hat{V}_r = \begin{pmatrix} \text{proportion of total standardized} \\ \text{sample variance in second set} \\ \text{explained by } \hat{V}_1, \hat{V}_2, \ldots, \hat{V}_r \end{pmatrix}$$

$$= \frac{\operatorname{tr}\left(\hat{\mathbf{b}}_{\mathbf{z}}^{(1)}\hat{\mathbf{b}}_{\mathbf{z}}^{(1),} + \cdots + \hat{\mathbf{b}}_{\mathbf{z}}^{(r)}\hat{\mathbf{b}}_{\mathbf{z}}^{(r)\,\mathsf{T}}\right)}{\operatorname{tr}\left(\mathbf{R}_{22}\right)}$$

$$= \frac{\sum_{i=1}^{r}\sum_{k=1}^{q} r^2_{\hat{V}_i, z_k^{(2)}}}{q}$$

Descriptive measures (10-37) provide some indication of how well the canonical variates represent their respective sets. In particular,

$$\frac{1}{p}\,\mathrm{tr}\left[\mathbf{R}_{11} - \hat{\mathbf{a}}_{\mathbf{z}}^{(1)}\hat{\mathbf{a}}_{\mathbf{z}}^{(1)\,\mathsf{T}} - \hat{\mathbf{a}}_{\mathbf{z}}^{(2)}\hat{\mathbf{a}}_{\mathbf{z}}^{(2)\,\mathsf{T}} - \cdots - \hat{\mathbf{a}}_{\mathbf{z}}^{(r)}\hat{\mathbf{a}}_{\mathbf{z}}^{(r)\,\mathsf{T}}\right]$$
$$= 1 - R^2_{\mathbf{z}^{(1)}}\,\hat{U}_1, \hat{U}_2, \ldots, \hat{U}_r$$
$$\frac{1}{q}\,\mathrm{tr}\left[\mathbf{R}_{22} - \hat{\mathbf{b}}_{\mathbf{z}}^{(1)}\hat{\mathbf{b}}_{\mathbf{z}}^{(1)\,\mathsf{T}} - \hat{\mathbf{b}}_{\mathbf{z}}^{(2)}\hat{\mathbf{b}}_{\mathbf{z}}^{(2)\,\mathsf{T}} - \cdots - \hat{\mathbf{b}}_{\mathbf{z}}^{(r)}\hat{\mathbf{b}}_{\mathbf{z}}^{(r)\,\mathsf{T}}\right]$$
$$= 1 - R^2_{\mathbf{z}^{(2)}}\,\hat{\hat{v}}_1, \hat{V}_2, \ldots, \hat{V}_r$$

# Example 10.7 (Calculating proportions of sample variance explained by canonical variates)

Consider the job characteristic-job satisfaction data discussed in Example 10.5. Using the table of sample correlation coefficients presented in that example, we find that

$$R^2_{\mathbf{z}^{(1)}|\hat{U}_1} = \frac{1}{5}\sum_{k=1}^{5} r^2_{\hat{U}_1, z_k^{(1)}} = \frac{1}{5}\left[(.83)^2 + (.74)^2 + \cdots + (.85)^2\right] = .58$$

$$R^2_{\mathbf{z}^{(2)}|\hat{V}_1} = \frac{1}{7}\sum_{k=1}^{7} r^2_{\hat{V}_1, z_k^{(2)}} = \frac{1}{7}\left[(.75)^2 + (.65)^2 + \cdots + (.50)^2\right] = .37$$

- The first sample canonical variate $\hat{U}_1$ of the job characteristics set accounts for 58% of the set's total sample variance. The first sample canonical variate $\hat{V}_1$ of the job satisfaction set explains 37% of the set's total sample variance.
- We might thus infer that $\hat{U}_1$ is a "better" representative of its set than $\hat{V}_1$ is of its set. The interested reader may wish to see how well $\hat{U}_1$ and $\hat{V}_1$ reproduce the correlation matrices $\mathbf{R}_{11}$ and $\mathbf{R}_{22}$, respectively.

When $\mathbf{\Sigma}_{12} = \mathbf{0}, \mathbf{a}^\top \mathbf{X}^{(1)}$ and $\mathbf{b}^\top \mathbf{X}^{(2)}$ have covariance $\mathbf{a}^\top \mathbf{\Sigma}_{12} \mathbf{b} = 0$ for all vectors $\mathbf{a}$ and b. Consequently, all the canonical correlations must be zero, and there is no point in pursuing a canonical correlation analysis. The next result provides a way of testing $\Sigma_{12} = \mathbf{0}$, for large samples.

# Result 10.3

Let $\mathbf{X}_j = \begin{bmatrix} \mathbf{X}_j^{(1)} \\ \mathbf{X}_j^{(2)} \end{bmatrix}$, $\quad j = 1, 2, \ldots, n$ be a random sample from an $N_{p+q}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ population with $\boldsymbol{\Sigma} = \begin{bmatrix} \underset{(p\times p)}{\boldsymbol{\Sigma}_{11}} & \underset{(p\times q)}{\boldsymbol{\Sigma}_{12}} \\ \underset{(q\times p)}{\boldsymbol{\Sigma}_{21}} & \underset{(q\times q)}{\boldsymbol{\Sigma}_{22}} \end{bmatrix}$ Then the likelihood ratio test of

$$H_0 : \Sigma_{12} = \underset{(p\times q)}{0} \quad \textit{versus} \quad H_1 : \Sigma_{12} \neq \underset{(p\times q)}{0}$$

rejects $H_0$ for large values of
$-2\ln\Lambda = n\ln\left(\frac{|\mathbf{S}_{11}||\mathbf{S}_{22}|}{|\mathbf{S}|}\right) = -n\ln\prod_{i=1}^{p}\left(1 - \widehat{\rho_i^{*2}}\right)$

where $\mathbf{S} = \begin{bmatrix} \mathbf{S}_{11} & \mathbf{S}_{12} \\ \mathbf{S}_{21} & \mathbf{S}_{22} \end{bmatrix}$ is the unbiased estimator of $\Sigma$. For large $n$, the test statistic $-2 \ln \Lambda$ is approx; distributed as a chi-square random variable with $pq$ d.f.

The likelihood ratio statistic compares the sample generalized variance: under $H_0$, namely,

$$\begin{vmatrix} \mathbf{S}_{11} & \mathbf{0} \\ \mathbf{0}^{\mathsf{T}} & \mathbf{S}_{22} \end{vmatrix} = |\mathbf{S}_{11}|\,|\mathbf{S}_{22}|$$

with the unrestricted generalized variance $|\mathbf{S}|$.

- Bartlett suggests replacing the multiplicative factor $n$ in the likelihood ratio statistic with the factor $n - 1 - \frac{1}{2}(p + q + 1)$ to improve the $\chi^2$ approximation to the sampling distribution of $-2 \ln \Lambda$.
- Thus, for $n$ and $n - (q + p)$ large, we reject $H_0 : \Sigma_{12} = 0 \left(\rho_1^* = \rho_2^* = \cdots = \rho_p^* = 0\right)$ at significance level $\alpha$ if

$$-\left(n - 1 - \frac{1}{2}(p + q + 1)\right) \ln \prod_{i=1}^{p} \left(1 - \widehat{\rho_i^*}^2\right) > \chi^2_{pq}(\alpha)$$

where $\chi^2_{pq}(\alpha)$ is the upper $(100\alpha)$ th percentile of $\chi^2_{pq}$

- If the null hypothesis $H_0 : \Sigma_{12} = 0 \left(\rho_1^* = \rho_2^* = \cdots = \rho_p^* = 0\right)$ is rejected, it is natural to examine the "significance" of the individual canonical correlations.
- We begin assuming that the first canonical correlation is nonzero and the remaining $p-1$ canonical correlations are zero. If this hypothesis is rejected, we assume that the first two canonical correlations are nonzero, but the remaining $p-2$ canonical correlations are zero, and so forth.
- Sequence of hypotheses are

$$H_0^k : \rho_1^* \neq 0, \rho_2^* \neq 0, \ldots, \rho_k^* \neq 0, \rho_{k+1}^* = \cdots = \rho_p^* = 0$$
$$H_1^k : \rho_i^* \neq 0, \text{ for some } i \geq k+1$$

- Reject $H_0^{(k)}$ at significance level $\alpha$ if

$$-\left(n-1-\frac{1}{2}(p+q+1)\right)\ln\prod_{i=k+1}^{p}\left(1-\widehat{\rho_i^*}^2\right) > \chi^2_{(p-k)(q-k)}(\alpha)$$

- $\chi^2_{(p-k)(q-k)}(\alpha)$ is the upper $(100\alpha)$ th percentile of a chi-square distribution with $(p-k)(q-k)$ d.f.
- $\prod_{i=k+1}^{p}\left(1-\widehat{\rho_i^*}^2\right)$ is the "residual" after the first $k$ sample canonical correlations have been removed from the total criterion $\Lambda^{2/n}=\prod_{i=1}^{p}\left(1-\widehat{\rho_i^*}^2\right)$.

# Some negative sides

- If the members of the sequence $H_0, H_0^{(1)}, H_0^{(2)}$, and so forth, are tested one at a time until $H_0^{(k)}$ is not rejected for some $k$, the overall significance level is not $\alpha$ and, in fact, would be difficult to determine.
- Another defect of this procedure is the tendency it induces to conclude that a null hypothesis is correct simply because it is not rejected.
- the overall test of significance is useful for multivariate normal data.